2011 Team Answers

T-1. $\frac{15}{4}$

T-2. 253

T-3. 20

T-4. $\frac{5\pi}{4}\sqrt{17}$

T-5. $\frac{9}{2}$

T-6. 96

T-7. 4096

T-8. 70

T-9. 11

T-10. $\frac{5}{4}$

2011 Team Solutions

T-1. The common ratio in the geometric sequence $1, x, y$ is $\frac{x}{1} = x$, so $y = x^2$. The arithmetic sequence $x, y, 3$ has a common difference, so $y - x = 3 - y$. Substituting $y = x^2$ in the equation yields

$$\begin{aligned} x^2 - x &= 3 - x^2 \\ 2x^2 - x - 3 &= 0, \end{aligned}$$

from which $x = \frac{3}{2}$ or -1. The respective values of y are $y = x^2 = \frac{9}{4}$ or 1. Thus the possible values of $x + y$ are $\frac{15}{4}$ and 0, so the answer is $\mathbf{\frac{15}{4}}$.

T-2. The requirement that a_n be the smallest positive value of $n - {a_k}^2$ for $k < n$ is equivalent to determining the largest value of a_k such that ${a_k}^2 < n$. For $n = 3$, use either $a_1 = a_2 = 1$ to find $a_3 = 3 - 1^2 = 2$. For $n = 4$, the strict inequality eliminates a_3, so $a_4 = 4 - 1^2 = 3$, but a_3 can be used to compute $a_5 = 5 - 2^2 = 1$. In fact, until $n = 10$, the largest allowable prior value of a_k is $a_3 = 2$, yielding the values $a_6 = 2$, $a_7 = 3$, $a_8 = 4$, $a_9 = 5$. In general, this pattern continues: from $n = m^2 + 1$ until $n = (m+1)^2$, the values of a_n increase from 1 to $2m+1$. Let $S_m = 1 + 2 + \cdots + (2m + 1)$. Then the problem reduces to computing $S_0 + S_1 + \cdots + S_6 + 1$, because $a_{49} = 49 - 6^2$ while $a_{50} = 50 - 7^2 = 1$. $S_m = \frac{(2m+1)(2m+2)}{2} = 2m^2 + 3m + 1$, so

$$\begin{aligned} S_0 + S_1 + S_2 + S_3 + S_4 + S_5 + S_6 &= 1 + 6 + 15 + 28 + 45 + 66 + 91 \\ &= 252. \end{aligned}$$

Therefore the desired sum is $252 + 1 = \mathbf{253}$.

T-3. Write $253_b \cdot 341_b = (2b^2 + 5b + 3)(3b^2 + 4b + 1) = 6b^4 + 23b^3 + 31b^2 + 17b + 3$. Compare the coefficients in this polynomial to the digits in the numeral $\underline{7}\ \underline{4}\ \underline{X}\ \underline{Y}\ \underline{Z}$. In the polynomial, the coefficient of b^4 is 6, so there must be a carry from the b^3 place to get the $7b^4$ in the numeral. After the carry, there should be no more than 4 left for the coefficient of b^3 as only one b is carried. Therefore $23 - b \leq 4$ or $b \geq 19$. By comparing digits, note that $Z = 3$. Then

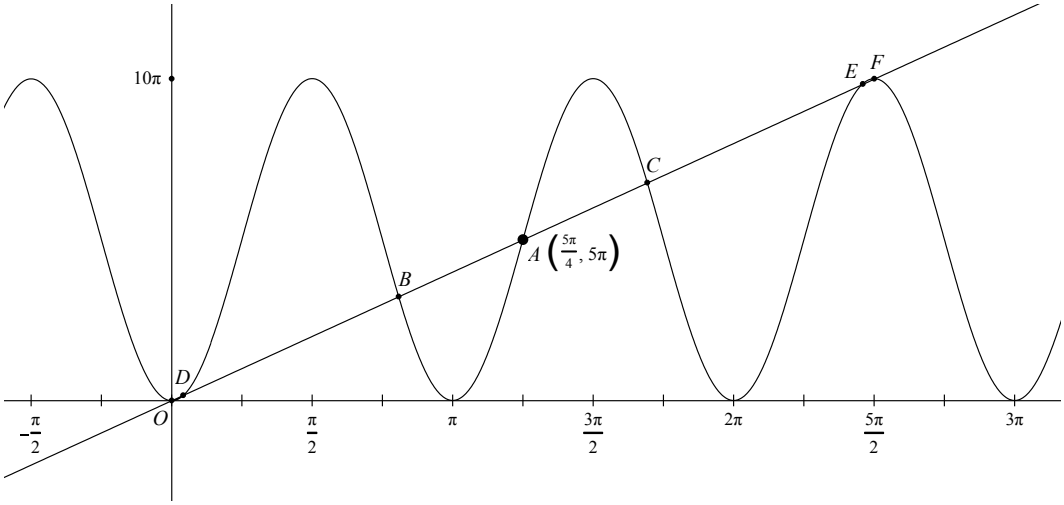
$$\begin{aligned} 6b^4 + 23b^3 + 31b^2 + 17b &= \underline{7}\ \underline{4}\ \underline{X}\ \underline{Y}\ \underline{0} \\ &= 7b^4 + 4b^3 + X \cdot b^2 + Y \cdot b. \end{aligned}$$

Because $b > 0$, this equation can be simplified to

$$b^3 + X \cdot b + Y = 19b^2 + 31b + 17.$$

Thus $Y = 17$ and $b^2 + X = 19b + 31$, from which $b(b - 19) = 31 - X$. The expression on the left side is positive (because $b > 19$) and the expression on the right side is at most 31 (because $X > 0$), so the only possible solution is $b = 20$, $X = 11$. The answer is **20**.

T-4. Notice first that all intersections of the two graphs occur in the interval $0 \le x \le \frac{5\pi}{2}$, because the maximum value of $10\pi \sin^2 x$ is 10π (at odd multiples of $\frac{\pi}{2}$), and $4x > 10\pi$ when $x > \frac{5\pi}{2}$. The graphs are shown below.



Within that interval, both graphs are symmetric about the point $A = \left(\frac{5\pi}{4}, 5\pi\right)$. For the case of $y = 10\pi \sin^2 x$, this symmetry can be seen by using the power-reducing identity $\sin^2 x = \frac{1-\cos 2x}{2}$. Then the equation becomes $y = 5\pi - 5\pi \cos 2x$, which has amplitude 5π about the line $y = 5\pi$, and which crosses the line $y = 5\pi$ for $x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \ldots$. Label the points of intersection A, B, C, D, E, F, and O as shown. Then $\overline{AB} \cong \overline{AC}$, $\overline{BD} \cong \overline{CE}$, and $\overline{OD} \cong \overline{EF}$. Thus

$$\begin{aligned} BD + AC + EF &= OD + DB + BA \\ &= OA. \end{aligned}$$

By the Pythagorean Theorem,

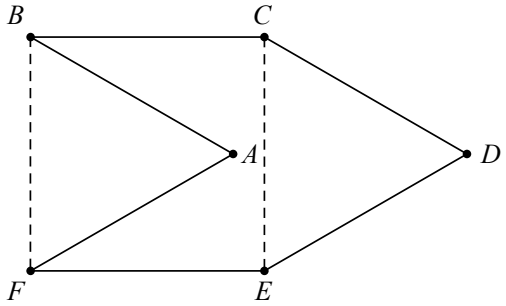
$$\begin{aligned} OA &= \sqrt{\left(\frac{5\pi}{4}\right)^2 + (5\pi)^2} \\ &= \frac{5\pi}{4}\sqrt{1^2 + 4^2} \\ &= \boxed{\frac{5\pi}{4}\sqrt{17}}. \end{aligned}$$

T-5. Let $\mathrm{m}\angle B = \alpha$. Then the sum of the measures of the angles in the hexagon is:

$$\begin{aligned} 720^\circ &= \mathrm{m}\angle A + \mathrm{m}\angle C + \mathrm{m}\angle E + \mathrm{m}\angle D + \mathrm{m}\angle B + \mathrm{m}\angle F \\ &= 10\alpha + 5\alpha + 5\alpha + 2\alpha + \alpha + \alpha = 24\alpha. \end{aligned}$$

Thus $30^\circ = \alpha$ and $\text{m}\angle A = 300^\circ$, so the exterior angle at A has measure $60^\circ = \text{m}\angle D$. Further, because $AB = CD$ and $DE = AF$, it follows that $\triangle CDE \cong \triangle BAF$. Thus

$$[ABCDEF] = [ABCEF] + [CDE] = [ABCEF] + [ABF] = [BCEF].$$



To compute $[BCEF]$, notice that because $\text{m}\angle D = 60^\circ$, $\triangle CDE$ is equilateral. In addition,

$$\begin{aligned} 150^\circ &= \text{m}\angle BCD \\ &= \text{m}\angle BCE + \text{m}\angle DCE = \text{m}\angle BCE + 60^\circ. \end{aligned}$$

Therefore $\text{m}\angle BCE = 90^\circ$. Similarly, because the hexagon is symmetric, $\text{m}\angle CEF = 90^\circ$, so quadrilateral $BCEF$ is actually a square with side length 3. Thus $CE = \frac{BE}{\sqrt{2}} = \frac{3}{\sqrt{2}}$, and $[ABCDEF] = [BCEF] = \mathbf{\frac{9}{2}}$.

Alternate Solution: Calculate the angles of the hexagon as in the first solution. Then proceed as follows.

First, $ABCDEF$ can be partitioned into four congruent triangles. Because the hexagon is equilateral and $\text{m}\angle ABC = \text{m}\angle AFE = 30^\circ$, it follows that $\triangle ABC$ and $\triangle AFE$ are congruent isosceles triangles whose base angles measure 75°. Next, $\text{m}\angle ABC + \text{m}\angle BCD = 30^\circ + 150^\circ = 180^\circ$, so $\overline{AB} \parallel \overline{CD}$. Because these two segments are also congruent, quadrilateral $ABCD$ is a parallelogram. In particular, $\triangle CDA \cong \triangle ABC$. Similarly, $\triangle EDA \cong \triangle AFE$.

Now let $a = AC = AE$ be the length of the base of these isosceles triangles, and let $b = AB$ be the length of the other sides (or of the equilateral hexagon). Because the four triangles are congruent, $[ABCDEF] = [ABC] + [ACD] + [ADE] + [AEF] = 4[ABC] = 4 \cdot \frac{1}{2} b^2 \sin 30^\circ = b^2$. Applying the Law of Cosines to $\triangle ABC$ gives $a^2 = b^2 + b^2 - 2b^2 \cos 30^\circ = (2 - \sqrt{3})b^2$. Because $4 - 2\sqrt{3} = (\sqrt{3} - 1)^2$, this gives $a = \left(\frac{\sqrt{3}-1}{\sqrt{2}}\right) b$. Using the given length $BE = 3$ and applying the Law of Cosines to $\triangle ABE$ gives

$$\begin{aligned} 9 &= a^2 + b^2 - 2ab\cos 135^\circ \\ &= a^2 + b^2 + \sqrt{2}ab \\ &= (2 - \sqrt{3})b^2 + b^2 + (\sqrt{3} - 1)b^2 \\ &= 2b^2. \end{aligned}$$

Thus $[ABCDEF] = b^2 = \mathbf{\frac{9}{2}}$.

T-6. Let $A = (0, 5)$ and $B = (12, 0)$, and let $C = (1, -1)$. First compute the distance sum: $d(A, C) + d(B, C) = 19$. Notice that if $P = (x, y)$ is on the segment from $(0, -1)$ to $(12, -1)$, then $d(A, P) + d(B, P)$ is constant. This is because if $0 < x < 12$,

$$\begin{aligned} d(A, P) + d(B, P) &= |0 - x| + |5 - (-1)| + |12 - x| + |0 - (-1)| \\ &= x + 6 + (12 - x) + 1 \\ &= 19. \end{aligned}$$

Similarly, $d(A, P) + d(P, B) = 19$ whenever P is on the segment from $(0, 6)$ to $(12, 6)$. If P is on the segment from $(13, 0)$ to $(13, 5)$, then P's coordinates are $(13, y)$, with $0 \le y \le 5$, and thus

$$\begin{aligned} d(A, P) + d(B, P) &= |0 - 13| + |5 - y| + |12 - 13| + |0 - y| \\ &= 13 + (5 - y) + 1 + y \\ &= 19. \end{aligned}$$

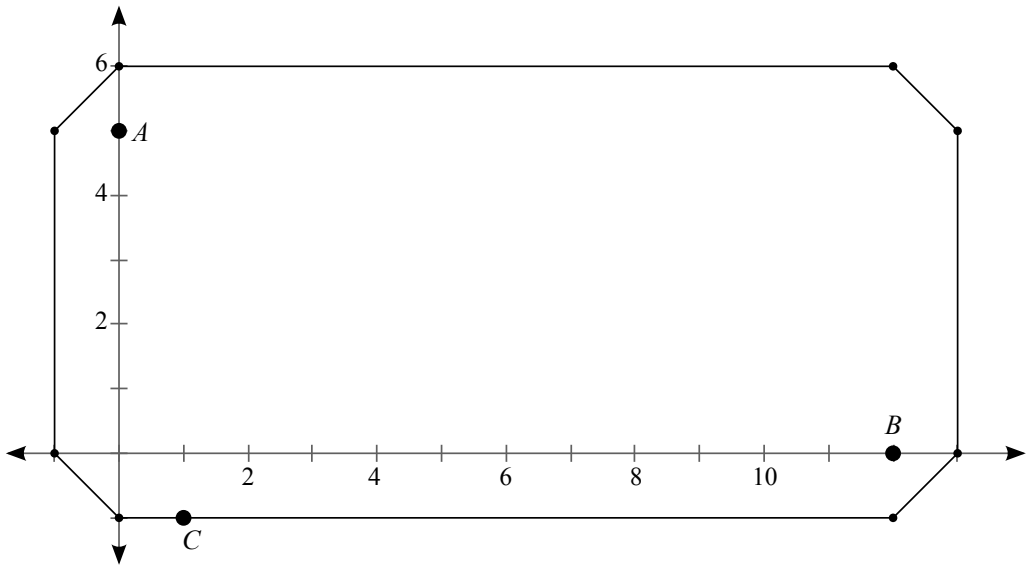
Similarly, $d(A, P) + d(P, B) = 19$ whenever P is on the segment from $(-1, 0)$ to $(-1, 5)$. Finally, if P is on the segment from $(12, -1)$ to $(13, 0)$, then $d(A, P) + d(B, P)$ is constant:

$$\begin{aligned} d(A, P) + d(B, P) &= |0 - x| + |5 - y| + |12 - x| + |0 - y| \\ &= x + (5 - y) + (x - 12) + (-y) \\ &= 2x - 2y - 7, \end{aligned}$$

and because the line segment has equation $x - y = 13$, this expression reduces to

$$\begin{aligned} d(A, P) + d(B, P) &= 2(x - y) - 7 \\ &= 2(13) - 7 \\ &= 19. \end{aligned}$$

Similarly, $d(A, P) + d(B, P) = 19$ on the segments joining $(13, 5)$ and $(12, 6)$, $(0, 6)$ and $(-1, 5)$, and $(-1, 0)$ to $(0, -1)$. The shape of the "ellipse" is given below.



The simplest way to compute the polygon's area is to subtract the areas of the four corner triangles from that of the enclosing rectangle. The enclosing rectangle's area is $14 \cdot 7 = 98$, while each triangle has area $\frac{1}{2} \cdot 1 \cdot 1 = \frac{1}{2}$. Thus the area is $98 - 4 \cdot \frac{1}{2} = \mathbf{96}$.

T-7. Substituting yields $\frac{512}{f(2)} = 2f(2) \Rightarrow (f(2))^2 = 256 \Rightarrow f(2) = 16$. Therefore $f(1) = \frac{1}{32}$. Using the recursion, $f(3) = \frac{1}{2}$ and $f(4) = 8$. So $f(f(4)) = f(8)$. Continue to apply the recursion:

$$f(5) = 4, \quad f(6) = 32, \quad f(7) = 128, \quad f(8) = \mathbf{4096}.$$

Alternate Solution: Let $g(n) = \log_2 f(n)$. Then $g(n) = g(n-1) + g(n-2)$, with initial conditions $g(1) = g(2) - 9$ and $-g(1) = 1 + g(2)$. From this, $g(1) = -5$ and $g(2) = 4$, and from the recursion,

$$g(3) = -1, \quad g(4) = 3,$$

so $f(4) = 2^{g(4)} = 8$. Continue to apply the recursion:

$$g(5) = 2, \quad g(6) = 5, \quad g(7) = 7, \quad g(8) = 12.$$

Because $g(f(4)) = 12$, it follows that $f(f(4)) = 2^{12} = \mathbf{4096}$.

T-8. When the coefficients of a polynomial f are reversed to form a new polynomial g, the zeros of g are the reciprocals of the zeros of f: r is a zero of f if and only if r^{-1} is a zero of g. In this case, the two polynomials have the *same* zeros; that is, whenever r is a zero of either, so must be r^{-1}. Furthermore, both r and r^{-1} must be real as well as integers, so $r = \pm 1$. As the only zeros are ± 1, and the greatest common divisor of all the coefficients is 1, the polynomial must have leading coefficient 1 or -1. Thus

$$\begin{aligned} f(x) &= \pm(x \pm 1)(x \pm 1) \cdots (x \pm 1) \\ &= \pm(x+1)^k (x-1)^{n-k}. \end{aligned}$$

If A_n is the number of such degree n polynomials, then there are $n+1$ choices for k, $0 \le k \le n$. Thus $A_n = 2(n+1)$. The number of such degree n polynomials for $1 \le n \le 7$ is the sum:

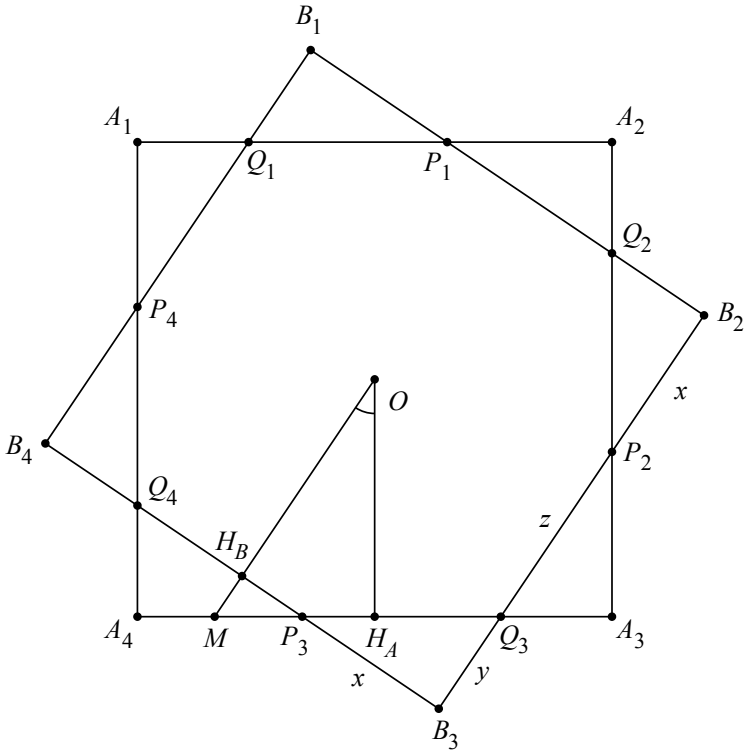
$$A_1 + A_2 + \ldots + A_7 = 2(2 + 3 + \ldots + 8) = 2 \cdot 35 = \mathbf{70}.$$

T-9. Label the sides of the polygon, in order, $s_0, s_1, \ldots, s_{15}$. First note that two sides of the polygon intersect at a vertex if and only if the sides are adjacent. So the sides chosen must be nonconsecutive. Second, if nonparallel sides s_i and s_j are extended, the angle of intersection is determined by $|i-j|$, as are the lengths of the extended portions of the segments. In other words, the *spacing* of the extended sides completely determines the shape of the triangle. So the problem reduces to selecting appropriate spacings, that is, finding integers $a, b, c \geq 2$ whose sum is 16. However, diametrically opposite sides are parallel, so (for example) the sides s_3 and s_{11} cannot both be used. Thus none of a, b, c may equal 8. Taking s_0 as the first side, the second side would be $s_{0+a} = s_a$, and the third side would be s_{a+b}, with c sides between s_{a+b} and s_0. To eliminate reflections and rotations, specify additionally that $a \geq b \geq c$. The allowable partitions are in the table below.

a	b	c	triangle
12	2	2	$s_0s_{12}s_{14}$
11	3	2	$s_0s_{11}s_{14}$
10	4	2	$s_0s_{10}s_{14}$
10	3	3	$s_0s_{10}s_{13}$
9	5	2	$s_0s_9s_{14}$
9	4	3	$s_0s_9s_{13}$
7	7	2	$s_0s_7s_{14}$
7	6	3	$s_0s_7s_{13}$
7	5	4	$s_0s_7s_{12}$
6	6	4	$s_0s_6s_{12}$
6	5	5	$s_0s_6s_{11}$

Thus there are **11** distinct such triangles.

T-10. In the diagram below, O is the center of both squares $A_1A_2A_3A_4$ and $B_1B_2B_3B_4$. Let P_1, P_2, P_3, P_4 and Q_1, Q_2, Q_3, Q_4 be the intersections of the sides of the squares as shown. Let H_A be on $\overline{A_3A_4}$ so that $\angle A_3H_AO$ is right. Similarly, let H_B be on $\overline{B_3B_4}$ such that $\angle B_3H_BO$ is right. Then the angle by which $B_1B_2B_3B_4$ was rotated is $\angle H_AOH_B$. Extend $\overline{OH_B}$ to meet $\overline{A_3A_4}$ at M.



Both $\triangle H_AOM$ and $\triangle H_BP_3M$ are right triangles sharing acute $\angle M$, so $\triangle H_AOM \sim \triangle H_BP_3M$. By an analogous argument, both triangles are similar to $\triangle B_3P_3Q_3$. Thus $\text{m}\angle Q_3P_3B_3 = \theta$. Now let $B_3P_3 = x, B_3Q_3 = y$, and $P_3Q_3 = z$. By symmetry, notice that $B_3P_3 = B_2P_2$ and that $P_3Q_3 = P_2Q_3$. Thus

$$x + y + z = B_3Q_3 + Q_3P_2 + P_2B_2 = B_2B_3 = 3.$$

By the Pythagorean Theorem, $x^2 + y^2 = z^2$. Therefore

$$\begin{aligned} x + y &= 3 - z \\ x^2 + y^2 + 2xy &= 9 - 6z + z^2 \\ 2xy &= 9 - 6z. \end{aligned}$$

The value of xy can be determined from the areas of the four triangles $\triangle B_iP_iQ_i$. By symmetry, these four triangles are congruent to each other. Their total area is the area not in both squares, i.e., $9 - 8 = 1$. Thus $\frac{xy}{2} = \frac{1}{4}$, so $2xy = 1$. Applying this result to the above equation,

$$\begin{aligned} 1 &= 9 - 6z \\ z &= \frac{4}{3}. \end{aligned}$$

The desired quantity is $\sin\theta + \cos\theta = \frac{x}{z} + \frac{y}{z}$, and

$$\begin{aligned} \frac{x}{z} + \frac{y}{z} &= \frac{x+y+z}{z} - \frac{z}{z} \\ &= \frac{3}{z} - 1 \\ &= \boxed{\frac{5}{4}}. \end{aligned}$$